

University System

Object-Oriented Programming & Design

Research-Oriented University Console System

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What Works

- Course registration with credit & fail count validation
- Mark system: Att1 + Att2 + Final → letter grade
- Transcript & GPA calculation
- Teacher: put marks, send complaint (LOW/MEDIUM/HIGH)
- Student organizations (member / head)
- Multilingual UI: KZ, EN, RU

- Research: h-index, printPapers, getCitation (BibTeX/Plain)
- GraduateStudent supervisor + LowHIndexException
- Observer pattern: journal subscription notifications
- Singleton (Database), Factory (UserFactory), Strategy (Comparators)
- Serialization: save & load system state
- TechSupport requests: VIEWED → ACCEPTED / REJECTED / DONE

Key Features Demo

Authentication

Login with email & password. Language auto-applies per user (KZ/EN/RU).

Course Registration

Validates MAX_CREDITS=21 and MAX_FAILS=3. SITE students can take Oil & Gas as free elective.

Put Marks

Teacher selects course → student → enters Att1, Att2, Final. Letter grade computed automatically.

Research Pipeline

Publish paper → auto-news generated → h-index recalculated → top-cited news pinned.

Observer Pattern

User subscribes to journal → notified instantly when new paper is added.

Serialization

System state saved to university.ser and restored on next run.

Limitations & What Does Not Work

No Schedule Generation

Automatic timetable generation taking room load and type into account is not implemented.

No Advanced Search

Regular expression-based search for students or courses is not implemented.

No Attendance Tracking

Lesson attendance recording for students is not supported.

Overall: Core OOP requirements fully satisfied — course registration, marks, research, patterns, serialization ✓